

C# Coding Interview Questions & Problems

Complete Reference Guide

SECTION 1: String Manipulation

Q1. Reverse Characters in a String

- a. Prepend Approach
- b. String Buffer (StringBuilder)
- c. Swap (in-place two-pointer)

Q2. Reverse Words in a String

Q3. Find Occurrence of Each Character in a Given String

Q5. Check if a String is a Palindrome

Q7. Anagram Code – Check if Two Strings are Anagrams

Q9. Find the First Non-Repeating Character

Q10. Remove Duplicates from a String

Q13. Extract the Surname from a Full Name String

Q14. Find the Longest Substring Without Repeating Characters

■ Example: $s = \text{"eghghhgg"} \rightarrow \text{Output: } 3 (\text{"egh"})$ | $s = \text{"substring"} \rightarrow \text{Output: } 8 (\text{"ubstring"})$

Q15. Remove All White Spaces from a String

Q16. Find the Longest Palindromic Substring

Q28. Longest Substring Without Repeating Characters (Sliding Window)

Write a function to return the length of the longest substring where all characters are distinct.

■ Input: $\text{"eghghhgg"} \rightarrow 3$ | $\text{"substring"} \rightarrow 8$

Q30. Valid Parentheses

Given a string of '(', ')', '{', '}', '[', ']', determine whether it is valid.

Rules: Every opening bracket must be closed by a matching type, and brackets must be closed in the correct order.

SECTION 2: Arrays & Numbers

Q4. Find Two Numbers in an Array That Add Up to a Target

■ Use a HashSet for $O(n)$ solution.

Q6. Reverse an Array

Q8. Move All 0's to the End of an Array

Q11. Swap Two Numbers (with and without temp variable)

Q12. Reverse a Number Without Converting to String

Q17. Print Array in Pattern

int[] GivenArray = { 1, 2, 3, 4, 5, 6 };

■ Output: //1 2 3 4//5 6 1 2

Q18. Left/Right Rotate an Array by K Places

Q21. Find the Maximum Number in an Array

Q27. Dutch National Flag / Sort Colors (0s, 1s, 2s)

Sort array nums in-place where values are 0, 1, 2 (red/white/blue).

Input: [2, 1, 2, 0, 1, 0, 1, 0, 1]

Output: [0, 0, 0, 1, 1, 1, 1, 2, 2]

Q29. Maximum Sum of a Contiguous Subarray of Size k (Sliding Window)

Given array nums and integer k, find the maximum sum of any contiguous subarray of size k.

■ Input: nums = [2,1,5,1,3,2], k=3 → Output: 9 (subarray [5,1,3])

SECTION 3: Math & Recursion

Q19. Prime Numbers – Check and List Primes

Q20. Factorial Using Recursion

Q22. Fibonacci Series

■ Output: 1, 1, 2, 3, 5, 8, 13, 21 ...

Q24. What Is the Next Number? Pattern: 1, 3, 7, 13, ...

■ Pattern: differences are 2, 4, 6, 8 → next = 21

SECTION 4: Patterns & Output

Q23. Basic Star Pattern Examples

Print various star/number patterns using nested loops.

Example patterns: Right triangle, Inverted triangle, Pyramid, Diamond

SECTION 5: SQL / Database

Q25. Find Duplicate Records in a Table

Use GROUP BY and HAVING COUNT(*) > 1.

Q26. Find Nth Highest Salary

Use ROW_NUMBER(), DENSE_RANK(), or nested subquery.

SECTION 6: Additional C# Interview Questions

Q31. OOP Concepts

Explain Encapsulation, Abstraction, Inheritance, Polymorphism with C# examples.

Q32. Abstract Class vs Interface

When to use abstract class vs interface in C#?

Q33. Value Types vs Reference Types

Difference between int/struct (value) and class/string (reference) types.

Q34. Boxing and Unboxing

What is boxing/unboxing? What are the performance implications?

Q35. Delegates, Events, and Lambda Expressions

Explain delegates and events. Write a simple event handler using lambda.

Q36. LINQ Queries

Write LINQ to filter, group, and order a list of objects.

Example: Find all employees with salary > 50000, grouped by department.

Q37. async / await and Task Parallel Library

Explain async/await. What is the difference between Task.Run and async?

Q38. Exception Handling Best Practices

try/catch/finally, custom exceptions, when NOT to swallow exceptions.

Q39. Garbage Collection & IDisposable

How does GC work in .NET? When should you implement IDisposable / using statement?

Q40. Collections: List, Dictionary, HashSet, Queue, Stack

When to use each? Time complexity of common operations.

Q41. String vs StringBuilder

Why use StringBuilder for concatenation in loops? Performance difference.

Q42. Nullable Types and Null Coalescing Operator

int? x = null; Use of ??, ??=, ?. operators.

Q43. Extension Methods

Write an extension method to check if a string is a palindrome.

Q44. Dependency Injection in .NET

Explain DI, IoC container, and how ASP.NET Core uses it.

Q45. Design Patterns in C#

Singleton, Factory, Repository, Observer – explain and implement one.

Q46. Generic Methods and Classes

Write a generic method Swap(ref T a, ref T b).

Q47. Reflection

What is reflection? How do you get type info at runtime?

Q48. Record Types (C# 9+)

Difference between class, struct, and record. When to use record?

Q49. Pattern Matching (switch expressions, is, when)

Use pattern matching to simplify type-checking code.

Q50. Memory Management: Stack vs Heap

How are local variables and objects stored? What causes a StackOverflowException?